Dark matter substructure: A lensing perspective

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The study of dark matter substructure through strong gravitational lensing has shown enormous promise in probing the properties of dark matter on subgalactic scales. This approach has already been used to place strong constraints on a wide range of dark matter models including self-interacting dark matter, fuzzy dark matter and warm dark matter. One major uncertainty in these studies is the strength of tidal stripping experienced by subhalos. We study theoretical predictions for the statistical properties of subhalos in strong gravitational lenses using the semianalytic galaxy formation toolkit GALACTICUS. We present a large suite of dark matter only GALACTICUS models, spanning nearly 2 orders of magnitude in host halo mass (from Milky Way to group mass halos between redshifts from 0.2 to 0.8). Additionally, to address potential shortcomings of including only dark matter physics we include a smaller set of GALACTICUS runs with the potential of a central massive elliptical to complement our dark matter only suite of models. We place particular focus on quantities relevant to strong gravitational lensing; namely the projected number density of substructure near the Einstein radius as a function of the host stellar mass and redshift. In the innermost region in projection, we find that our GALACTICUS models agree with N-body simulations within a factor of ~ 2 within the Einstein radius. We find that the addition of a central galaxy suppresses the projected number density of subhalos within the Einstein radius by around 15% relative to dark matter only simulations.

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I. INTRODUCTION

Cold dark matter (CDM) provides an excellent description of the matter distribution on the largest scales. For example, the CDM model has had tremendous success explaining the cosmic microwave background [1] and the mass function and spatial distribution of luminous galaxies (e.g., [2–5]). Recently, probes of dark matter have begun pushing the frontier to subgalactic scales without the need for luminous tracers. In this regime, stellar streams (e.g., [6–10]) promise to provide constraints within the local group. Outside the local group, strong gravitational lensing probes the properties of low mass halos (e.g., [11–20]).

One source of uncertainty in gravitational lensing analyses of dark matter is the normalization of the subhalo mass function (SHMF). Halos can be considered in two categories: (1) isolated (field) halos and (2) halos halo (subhalos). Both categories contribute to the lensing observables. While the properties of field halos can be robustly predicted for a given dark matter model, subhalos undergo complex interactions with the host including tidal heating, tidal stripping and dynamical friction, which all occur within the evolving potential of the host halo. These physical effects map the subhalos at infall (the so-called unevolved population) to the subhalos at a later time (referred to as the evolved population). An increase in tidal stripping results in a decrease in the normalization of the evolved SHMF. It is the evolved population of subhalos that have been affected by tidal interactions with the host that is observed during strong gravitational lensing measurements, whereas the unevolved population depends only on subhalo infall masses. Mass loss due to tidal stripping is spatially dependent [21] with some subhalos having evolved masses which are 1% or less of their infall masses [22]. In the case of a subhalo with 1% of its original infall mass remaining, the potential lensing signal (percent change in image magnification) can be reduced by a factor of ~ 4 [23] compared to the signal that would be observed absent tidal interactions.

The subhalo to field halo ratio is strongly dependent on the lens and source redshift. N-body simulations suggest that subhalos account for 5%–20% of the total number of halos near the lensed images in projection [24,25]. Although subhalos are subdominant in number, Gilman *et al.* [26] showed that uncertainty in tidal stripping is a major source of degeneracy in strong gravitational lensing studies. Current forward modeling pipelines do not have variable models of tidal stripping strength; instead, the normalization

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of the unevolved SHMF ($\Sigma_{\rm sub}$) is allowed to vary. This serves as a proxy for varying tidal stripping strength. To study the impact of the $\Sigma_{\rm sub}$ prior on lensing inferences, Gilman *et al.* [26] simulated 25 mock lenses with warm dark matter with a half-mode mass ($m_{\rm hm} = 10^{7.5} M_\odot$) as a ground truth. Gilman *et al.* [26] compared posteriors obtained using a 0.2 dex Gaussian prior on $\Sigma_{\rm sub}$ with to that obtained using a uniform prior of 1.5 dex. Changing the prior from the wide uniform prior to the narrow Gaussian prior lowered the bound on the half-mode-mass by 2 dex. We aim to investigate whether a narrower prior on $\Sigma_{\rm sub}$ can be used given current theoretical uncertainties.

We highlight that caution must be exercised: A large body of work has drawn attention to artificial disruption present in N-body simulations [27–30]. Even in high resolution simulations, such as the Caterpillar [31] suite of N-body simulations of Milky Way mass halos with a per particle mass of $\sim 10^4 M \odot$ artificial disruption can lead to a 10%–20% suppression of the SHMF over the entire viral volume [30]. Additionally, Benson and Du [30] found a significant spatial dependence in the effects of artificial disruption, with the SHMF in the inner 2% of the virial radius spuriously suppressed by nearly a factor of 3. Due to the geometry of projection, subhalos at small radial displacement from the host are overrepresented in the populations of subhalos probed by lensing, making the effects of artificial disruption more pronounced when compared to the total subhalo population.

Due to their computational cost, close-up simulations are not conducive to studying population level statistics of subhalos over the large range of parameter space of host halo masses and redshift representative of lenses used to study dark matter. Analytic and semianalytic models of subhalo interactions provide a path forward. These methods must account for the complex interactions between the subhalos and their host. Developing fast and accurate models of the impact of environment on subhalos remains an area of active study. In particular, models of tidal stripping have recently received a great deal of attention, with much recent work focused on understanding the tidal tracks of subhalo evolution [22,30,32,33].

In light of the improved understanding of artificial disruption in N-body simulations, this study revisits the predictions for the joint spatial and mass distribution of subhalos in CDM, with emphasis on quantities related to gravitational lensing. We use semianalytic models as our primary mode of emulating dark matter subhalos. In particular, we use the GALACTICUS[1] [34] galaxy formation toolkit. The GALACTICUS galaxy formation toolkit is a modular and open source semianalytic model of galaxy formation with an extensive library of both dark matter and baryonic physics.

[1]https://github.com/galacticusorg/galacticus; we use revisions 723170fb1690257be9b0588c0a83c9e559a584ae and 0e3a5adea6e3b32b9eaaa3a6c4896bbe63fb0cb1.

We present a large suite of GALACTICUS subhalo realizations, spanning nearly two decades in host halo mass (from Milky Way mass to group mass halos at redshifts 0.2–0.8). We use our suite of GALACTICUS runs to study quantities relevant to strong gravitational lensing, with particular focus placed on the subhalo population within a 20 kpc aperture. A 20 kpc aperture is chosen to be small enough to be comparable to the Einstein radius of a typical Sloan Lens Advanced Camera Survey (SLACS) lens [35] ($\sim$1 arc sec corresponding to $\sim$8 kpc for a lens at redshift $z = 0.5$), but large enough to obtain reasonable statistics from our GALACTICUS models. To complement our GALACTICUS models, we include a comparison with the Symphony suite of N-body simulations and the Han *et al.* [21] analytic model of the spatial and mass distribution of subhalos. Similar to previous works focusing on the subhalo populations of strong gravitational lenses, our main suite of models is limited to dark matter only physics. We make a preliminary study of baryonic effects by studying a smaller suite of models that include the potential of a central galaxy. We take an empirical approach to modeling the evolution of the central galaxy using the UNIVERSEMACHINE [36] correlation between galaxy growth and dark matter halo assembly.

In Sec. II, we discuss the parameters chosen for our GALACTICUS models and the methods used in our analysis. Next, in Sec. III we present our results and provide discussion in Sec. IV. Finally, we summarize our results in Sec. V. For this work, we assume cosmological parameters from the Planck Collaboration [1], $(H_0, \Omega_m, \Omega_\Lambda) = (67.36, 0.31530, 0.68470)$. A combination of ASTROPY [37–39], COLOSSUS [40] and GALACTICUS [34] software packages are used for cosmological calculations. To calculate halo concentrations the Diemer and Joyce [41] model is used. All quantities are reported in physical units unless otherwise specified.

II. METHODS

In this section we discuss the methods used in this work. We discuss our semianalytic models in Sec. II A. In Sec. II B we provide an overview of theoretical expectations for the spatial distribution and mass function of subhalos.

A. Semianalytic models

We utilize semianalytic models as our primary mode of studying the population level statistics of dark matter substructure. Semianalytic models have the advantage of speed when compared to N-body simulations, and are chosen to be the primary focus of this work to enable fast exploration of the population level statistics of the substructure of dark matter halos. In particular, we use GALACTICUS, an open source and extensible galaxy formation framework. The GALACTICUS toolkit is particularly well suited to the study of subhalo properties due to its

extensive library of subhalo physics and its recent calibration to high resolution idealized simulations provided in Du *et al.* [33].

Here, we give a brief summary of the algorithms used to generate realizations of dark matter substructure. We use GALACTICUS to generate merger trees and then subsequently evolve the trees forward in time using a set of analytic models for the evolution of subhalo density profiles and orbits (orbit initialization, dynamical friction, tidal stripping and tidal heating). In GALACTICUS, dark matter merger trees are built using Monte Carlo algorithms presented in Cole *et al.* [42] and Parkinson *et al.* [43]. In this algorithm, merger trees are evolved backward in time, with branching rates calculated at each using extensions [44,45] to Press and Schechter [46] formalism. Merger tree nodes are then evolved forward in time. First, orbits are initialized using the potential of the host halo [47]. After initialization, subhalo orbits are then evolved using models for dynamical friction [48], tidal stripping [49] and tidal heating [50].

1. Tidal physics

As a subhalo evolves within its host halo, it gradually loses mass due to tidal stripping; i.e., particles are removed over time by the tidal forces exerted by the host. In addition, tidal shocks heat the subhalo, causing the density profile of the subhalo to expand and become less dense, causing further mass loss. For massive subhalos (subhalos with subhalo-to-host halo mass ratios close to 1), the trajectories of host particles are bent by the gravity of subhalos, creating an overdensity behind the subhalo. This results in a net drag force known as dynamical friction, which causes the subhalo's orbital radius to decay and eventually leads to its merger with the host. As the subhalo moves closer to the host's center, it experiences increasingly stronger tidal forces, accelerating mass loss. In our work, we model subhalo evolution using the model described in Refs. [30,33,51] within the GALACTICUS framework.

We use subhalo physics tuned to high resolution idealized simulations provided by Du *et al.* [33] in which subhalos are evolved in an analytic host potential. Compared to tradition cosmological close-up simulations, these simulations have the advantage of resolving subhalos at much higher resolution for less computational cost. For example, in the idealized simulations used by Du *et al.* [33], subhalos start with $N \approx 10^7$ particles and are tracked until $N \approx 10^4$ particles remain. This is in contrast to high resolution cosmological simulations such as Caterpillar [31], where the lowest mass subhalos are tracked to $N \approx 20$ particles. The high resolution minimizes the effects of numerical issues traditionally associated with N-body simulations, such as artificial disruption [52]. Additionally, the challenges associated with halo finding in cosmological simulations is not present in idealized simulations. However, we note that our idealized simulations have several limitations; idealized simulations lack cosmological context and use simplified, spherically symmetric analytic profiles to describe the host halo and infalling subhalo.

2. Merger radius

Probes of strong gravitational lensing are sensitive to subhalos appearing in projection near lensed images which appear near the Einstein radius. For a typical group mass lens at redshift $z = 0.5$, this aperture has a radius of ∼1 arc sec (∼7 kpc) [35]. Due to effects of projection, strong lensing probes are especially sensitive to subhalos at small spatial separation to the host. The GALACTICUS galaxy formation model automatically destroys subhalos within a minimum radius. We select the minimum radius to simulate subhalos by running convergence tests. We find convergence for the spatial distribution at 10 kpc for a merging radius of $0.01 r_{\rm v}$. For a $10^{13.5} M_\odot$ mass halo at redshift, $z = 0.2$, $0.01 r_{\rm v}$ corresponds to a radial separation from the center of the host of 0.7 kpc. Unless otherwise noted, when considering projected quantities, we include subhalos within an annulus with an inner and outer radius of 10 and 20 kpc respectively. This ensures that the subhalo population within 10 kpc, which may be incomplete, is excluded. We run our models with a tree (infall mass) floor of $8 \times 10^7 M_\odot$. To ensure the completeness of the subhalo population, we track subhalos to arbitrarily low bound mass.

3. Host halo properties

We model across a broad range of masses and redshifts to capture the evolution of the projected number density and to facilitate comparison with observational works as well as other simulations. Our suite of GALACTICUS models consists of a ten by ten grid of GALACTICUS outputs spanning the range of host halo masses $12.0 \leq \log_{10}(M_{\rm h}/M_\odot) \leq 13.5$ and redshifts $0.2 \leq z \leq 0.8$ with 224 host halos per grid point.

4. Central galaxy properties

In addition to our dark matter only models, we include a smaller suite of GALACTICUS models spanning $13.0 \leq \log_{10}(M_{\rm h}/M_\odot) \leq 13.5$ with the inclusion of the potential of a central massive elliptical. Along with the increased tidal heating and stripping due to the potential of the central galaxy, we include a model of baryonic contraction due to the central potential [53]. The morphology of the central massive elliptical is modeled using a Hernquist [54] profile. We take care to accurately model the evolution of the stellar mass and scale radius of the central galaxy; the evolution of the central galaxy is modeled using the stellar mass to halo mass relation of Behroozi *et al.* [36], Eq. (J3). For our work, we use the best-fit parameters provided in the first row of Table J1. For simplicity, we do not include intrinsic scatter in the stellar mass to halo mass relation. We use empirical power law fits provided by Shen *et al.* [55] for the stellar mass to radius relationship for early-type galaxies in

the Sloan Digital Sky Survey [56] to evolve the Hernquist radius of the central galaxy. Similarly, we do not include scatter in this relationship. We provide further discussion of the evolution of the central galaxy in Appendix.

B. Analytic models

Here, we discuss theoretical expectations for the subhalo population. We place particular emphasis on the spatial and mass distribution of subhalos. Two cases of the spatial-mass distribution can be considered, using two definitions of subhalo mass: the mass at infall and the gravitationally bound mass of the subhalo at the time of observation. The former is known as the "unevolved" distribution and the latter the "evolved" distribution. By definition, no mass loss due to tidal stripping is considered in the unevolved distribution, while the evolved distribution includes tidal mass loss. Due to the extreme mass loss undergone by some subhalos, the unevolved distribution is not an observable quantity. Instead, the unevolved distribution provides a powerful tool to study the properties of subhalos, allowing the complicated physics of tidal stripping to be separated from orbital physics. A useful picture is that the evolved distribution can be approximately thought of as being derived from the unevolved distribution and a model of tidal mass loss [21,23]. However, we note that this picture is not exact. For example, physics of tidal stripping cannot be completely separated from orbital physics as tidal mass loss can affect dynamical friction (GALACTICUS uses the bound mass in dynamical friction calculations).

We compare the predictions from N-body simulations and semianalytic models with the analytic prescription presented by Han *et al.* [21]. Here, we describe the Han *et al.* [21] model and discuss some of its physical assumptions. Han *et al.* [21] first assume that the unevolved spatial-mass distribution of subhalos is separable (the spatial and mass function can be separated into a mass function and spatial distribution with the mass function only dependent on the subhalo's mass and the spatial distribution only dependent on the subhalo's radial separation from the host). Then Han *et al.* [21] combine this assumption with the assumption that the unevolved spatial distribution of subhalos traces the density profile of the host to arrive at the unevolved spatial and mass distribution:

$$\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m} \propto \tilde{\rho}(r)\frac{\mathrm{d}N}{\mathrm{d}m}(m), \tag{1}$$

where $\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m}$ is the unevolved subhalo spatial and mass function, $\tilde{\rho}$ is the normalized density profile of the host, $\mathrm{d}N/\mathrm{d}m$ is the unevolved mass function and m is the unevolved (infall) mass of the subhalo. The unevolved mass function is then approximated as a power law

$$\frac{dN}{dm} \propto m^{\alpha}, \tag{2}$$

with $\alpha \sim -2$ being the logarithmic slope of the subhalo mass function. Next, Han *et al.* [21] assume that the gravitationally bound (evolved) mass of a subhalo can be related to the unevolved mass of the subhalo on average using a transfer function dependent only on the distance from the center of the host

$$\frac{m_b}{m} = T(r), \tag{3}$$

where $T(r)$ is the spatially dependent transfer function and m_b is the evolved mass of the subhalo. In their analysis, Han *et al.* [21] take $T(r)$ to be a power law

$$T(r) \propto \left(\frac{r}{r_{\mathrm{v}}}\right)^{\beta}, \tag{4}$$

where $\beta \sim 1$ is a free parameter.

Plugging Eqs. (2) and (3) into Eq. (1) gives the evolved spatial and mass function

$$\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m_{\mathrm{b}}} \propto \hat{T}(r)\tilde{\rho}(r)m_b^{\alpha}, \tag{5}$$

where $\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m_{\mathrm{b}}}$ is the evolved spatial and mass function and $\hat{T}(r) = (T(r))^{-(\alpha+1)}$ is substituted for brevity. We note that for $\alpha \sim -2$, $\hat{T}(r) \approx T(r)$. Furthermore, substituting the form of $T(r)$ used by Han *et al.* [21] [Eq. (4)] gives

$$\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m_{\mathrm{b}}} \propto r^{\gamma}\tilde{\rho}(r)m_b^{\alpha}, \tag{6}$$

where $\gamma = -\beta(\alpha+1) \sim 1$. Han *et al.* [21] fit γ to various N-body simulations finding $\gamma = 0.95$ for Milky Way mass Aquarius halos [57] and $\gamma = 1.33$ for group mass halos in the Phoenix suite [58], indicating a dependence on host halo mass.

To analyze our GALACTICUS results, we fit the Han *et al.* [21] models to the GALACTICUS predictions for the spatial and mass function in Sec. III. To provide an additional point of comparison to our GALACTICUS results in the context of gravitational lensing, we spatially project the Han *et al.* [21] into 2D. An immediate consequence of the separability assumed by the Han *et al.* [21] model is a prediction that the logarithmic slope of the SHMF should be identical between the projected and unprojected distributions. Additionally, we study the predictions of the model for the scaling of the projected subhalo mass function (PSHMF) within the Einstein radius, an important quantity for gravitational lensing. We estimate the scaling of the projected spatial and mass function as a function of the host halo mass utilizing the host halo mass scaling relations discussed in Han *et al.* [21]. However, redshift-dependent scaling is not considered by Han *et al.* [21], who derived their model for a single snapshot in time at $z = 0$. Typical lensing galaxies

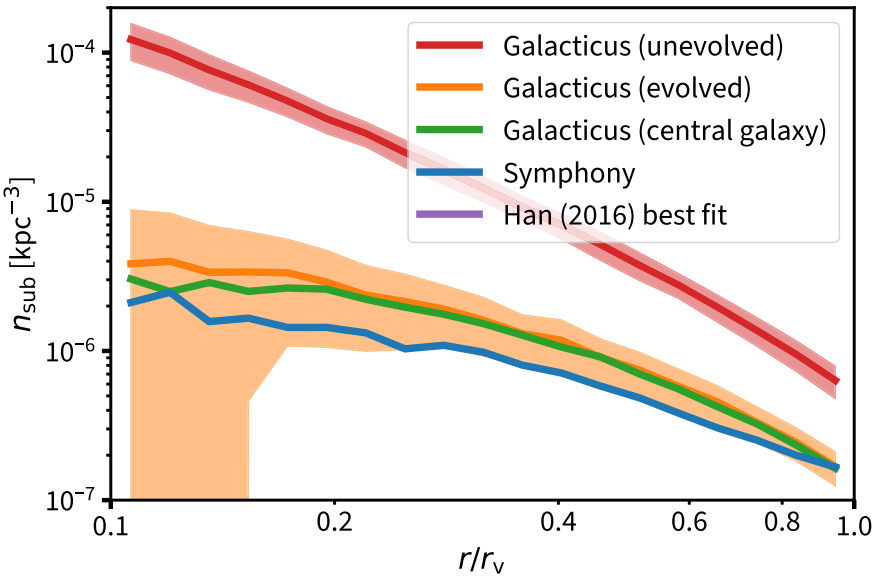


FIG. 1. The differential spatial distributions ($n_{\rm sub}$) from our GALACTICUS models and Symphony for subhalos with mass $9 < \log_{10}(m_b/M_\odot) < 10$ plotted as a function of the virial radius fraction ($r/r_{\rm v}$). Results are for halos with mass $M_{\rm h} = 10^{13} M_\odot$ at redshift $z = 0.5$ In red: the unevolved spatial distribution from our GALACTICUS models. In orange: the evolved spatial distribution for GALACTICUS and in green with a central galaxy. The GALACTICUS and Symphony spatial distributions shown are a result of averaging over the spatial distributions of each individual halo in the suite at each bin over all halos in the suite. The halo-to-halo scatter is calculated in a similar way, taking the standard deviation of the values of the spatial distributions of each of the host halos calculated at each radial bin. The shaded regions show the 1σ halo-to-halo scatter for the GALACTICUS evolved and unevolved cases; scatter for the other distributions is not shown for visual clarity. The best fit of the Han (2016) model to the GALACTICUS DMO unevolved spatial distribution is shown in purple. In blue: the spatial distribution for Symphony. We find there is no significant difference in the spatial distribution of subhalos when running models with or without a central galaxy.

are found at redshifts $0.2 \lesssim z \lesssim 0.8$, so understanding the scaling of the SHMF with redshift is essential. Therefore, we discuss extending the Han *et al.* [21] model over this range of redshifts in the following sections.

III. RESULTS

Here, we analyze our suite of GALACTICUS models. In Sec. III A we discuss the spatial distribution predicted by GALACTICUS. Next, in Sec. III B we calculate the normalization of the PSHMF of our dark matter only (DMO) simulations, and repeat the calculations for our models with the potential of a central massive elliptical in Sec. III C. Next, in Sec. III D we predict the scaling of the PSHMF as a function of host halo mass and redshift using our dark matter simulations. Finally, we discuss the impact of a central galactic potential on the scaling of the PSHMF.

A. Spatial distribution

Figure 1 shows the spatial distribution predicted by GALACTICUS for a $10^{13} M_\odot$ halo at $z = 0.5$, with both the evolved and unevolved distributions plotted. In addition to the GALACTICUS predictions, results are shown for Symphony [59] and the Han *et al.* [21] model of the spatial distribution. We plot the ratio of the radial distribution of subhalos to the host's dark matter density in Fig. 2. For a $10^{13} M_\odot$ halo at $z = 0.5$ we find the best fit to the GALACTICUS spatial distribution with a value of $\gamma = 0.98$ (shown in Table I). Additionally, fits provided by Han *et al.* [21] to Aquarius [57] and Phoenix [58] are included for comparison in Table I.

The fits to the $10^{12.0} M_\odot$ GALACTICUS halos agree well with fits to Aquarius halos ($\approx 10^{12.0} M_\odot$). We find a trend of decreasing γ with host halo mass in our suite of GALACTICUS models. This is opposite of the trend of increasing γ with halo mass found by Han *et al.* [21] when comparing the Aquarius and Phoenix simulations ($\approx 7 \times 10^{14} M_\odot$). Over the range of redshifts probed $0.2 \leq z \leq 0.8$, and at a fixed halo mass we find that γ is nearly constant. Further discussion of the Han *et al.* [21] model is provided in Sec. IV A.

Figure 3 shows the mean and standard deviation of the projected spatial distributions predicted by GALACTICUS for subhalos in the mass range $8 < \log_{10}(m/M_\odot) < 9$ for a $10^{13} M_\odot$ halo at $z = 0.5$ with projected radii $r_{\rm 2d} < 20$ kpc. The mean and standard deviation are shown for each 76kpc^2 radial bin, combining an aggregate of three projected spatial distributions (in the xy, yz, zy planes) per host halo.

For reference, we show the median Einstein radius from the sample of SLACS lenses presented by Shu *et al.* [35]. We find that the projected spatial distribution

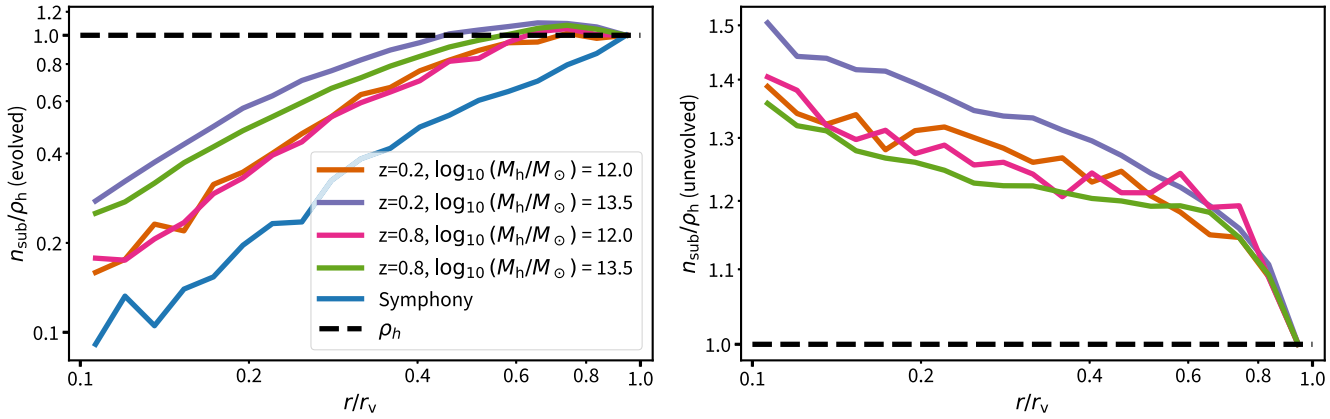


FIG. 2. The ratio of the spatial distribution of subhalos ($n_{\rm sub}$) to the host's density profile ($\rho_{\rm h}$) for subhalos with $8 < \log_{10}(m_b/M_\odot) < 9$ normalized to 1 at $r = r_{\rm v}$ for the evolved distribution (left) and $8 < \log_{10}(m/M_\odot) < 9$ unevolved distribution (right). A horizontal line of 1 (black dotted) shows the host halos density profile. Results are plotted for GALACTICUS over a range of host halo masses and redshifts and for Symphony for a single mass and redshift [$\log_{10}(M_{\rm h}/M_\odot) = 13, z = 0.5$]. The relative decrease toward the center in the evolved case is a result of tidal stripping (see Sec. II B), while the relative increase in the unevolved case is likely a result of dynamical friction (see Sec. IV A). Over our range of redshifts explored by our GALACTICUS simulations, the slope of the evolved spatial distribution $n_{\rm sub}/\rho_{\rm h}$ has a mild dependence on halo mass, with almost no dependence on redshift. For the unevolved case, there appears to be no significant dependence on either redshift or host halo mass.

is nearly constant in the innermost region of the host, which is consistent with the findings of Xu *et al.* [24]. We find that this holds true for both our DMO models and models with a central galaxy.

B. Projected subhalo mass function normalization: Dark matter only

Of particular interest to lensing studies is the integrated number of subhalos within an aperture near the Einstein radius as a function of the properties of the host halo. To measure this, we use the convention of Gilman *et al.* [18] to define the average density of the unevolved PSHMF within the Einstein radius as

$$\frac{{\rm d}^2N}{{\rm d}m{\rm d}A} = \frac{\Sigma_{\rm sub}}{m_0}\left(\frac{m}{m_0}\right)^{\alpha} F(M_{\rm halo}, z), \tag{7}$$

TABLE I. Table of best fits to the Han *et al.* [21] model of the spatial distribution [see Eq. (5)]. We include fits spanning the range of halo masses and redshifts included in our suite of GALACTICUS models, as well as at the redshift and host halo mass, $M_h = 10^{13}M_\odot$ and redshift, $z = 0.5$ where $\Sigma_{\rm sub}$ is defined. For comparison, we include fits to Aquarius and Phoenix halos provided by Han *et al.* [21].

Simulation	$\log_{10}(M_{\rm h}/M_\odot)$	z	γ
GALACTICUS	12.0	0.2	1.01
	12.0	0.8	1.06
	13.0	0.5	0.82
	13.5	0.2	0.67
	13.5	0.8	0.75
Aquarius	12.0	0.0	0.95
Phoenix	14.8	0.0	1.33

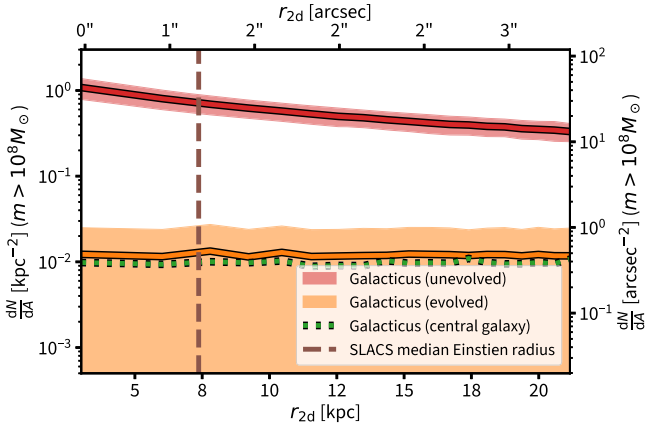


FIG. 3. The projected number density within the inner 20 kpc for both the unevolved (upper red), evolved with central galaxy (green dotted) and evolved (lower orange) subhalo distributions. Predictions from GALACTICUS are shown for subhalos with mass $>10^8 M_\odot$ in a $10^{13}M_\odot$ halo, with scatter shown for 76 kpc^2 bins. In brown (dotted) the Einstein radius for a typical SLACS lens is shown. The evolved number density is constant within the innermost region of the halo for both DMO and central galaxy models. The solid line shows the mean, while the shaded region shows 1σ halo-to-halo scatter (see Sec. III A for more details). Scatter is not shown for the case with a central galaxy for visual clarity. The projected spatial distribution with the inner 20 kpc is nearly independent of projected distance. Additionally, for our group scale simulations, we find that the addition of a central massive elliptical has a negligible impact on the projected number density of subhalos.

TABLE II. Table of best fits for the projected subhalo mass function [PSHMF, Eqs. (7) and (8)]. Fitting coefficients are averaged over 224 host halos per redshift and host halo mass (22,400 host halos in total). Projections are taken within a 10–20 kpc annulus, and averaged over three projections in the xy, yz, and xz planes. Due to the low number of halos in the aperture, halo-to-halo scatter for the SHMF slope cannot be calculated.

Parameters	Description	Value	Halo-to-halo scatter	Units
$\Sigma_{\rm sub}$	Unevolved PSHMF amplitude	420	73.5	10^{-3} kpc^{-2}
$f_{\rm s}$	PSHMF tidal stripping coefficient	0.0282	0.0105	⋯
$f_{\rm m}$	Average subhalo mass loss fraction	0.0252	0.0095	⋯
$f_{\rm s}\cdot\Sigma_{\rm sub}$	Evolved PSHMF amplitude	11.6	3.99	10^{-3} kpc^{-2}
α	Unevolved logarithmic PSHMF slope	−1.94	⋯	⋯
$\alpha_{\rm b}$	Evolved logarithmic PSHMF slope	−1.97	⋯	⋯

where $\frac{{\rm d}^2N}{{\rm d}m{\rm d}A}$ is a measure of the projected subhalo number density evaluated at the Einstein radius, m is the mass of a subhalo at infall, m_0 is a pivot mass taken by convention to be $10^8 M_\odot$, and the function $F(M_{\rm halo}, z)$ describes the dependence on host halo mass and redshift. For convenience, we refer to $\frac{{\rm d}^2N}{{\rm d}m{\rm d}A}$ as the projected subhalo mass function (instead of "projected subhalo mass density function"). We note that Eq. (7) can be derived by projecting Eq. (1). The scaling term $F(M_h, z)$ is a result of the dependence of the total number of subhalos on halo mass and redshift as well as the dependence of the spatial distribution of subhalos on both halo mass and redshift (for the unevolved case, Han *et al.* [21] models the spatial distribution of subhalos as following the density profile of the host). Factoring out all dependence on the redshift and host halo mass into the scaling function $F(M_{\rm halo}, z)$ allows a single PSHMF normalization ($\Sigma_{\rm sub}$) to be measured for all host halos regardless of host halo mass or redshift.

To provide further context for Eq. (7), we can define the evolved PSHMF:

$$\frac{{\rm d}^2N}{{\rm d}m_{\rm b}{\rm d}A} = \frac{f_s\cdot\Sigma_{\rm sub}}{m_0}\left(\frac{m_b}{m_0}\right)^{\alpha_{\rm b}} F_b(M_{\rm halo}, z), \tag{8}$$

where f_s is the reduction in normalization from the unevolved case, $\alpha_{\rm b}$ is the logarithmic slope of the evolved SHMF and $F_b(M_h, z)$ describes the halo mass/redshift scaling of the evolved PSHMF. Equation (8) can be derived using a similar procedure to that used by Han *et al.* [21]: Define an average ratio of subhalo bound mass to infall mass for those subhalos within the Einstein radius $m/m_b = f_m$ and project Eq. (5). This derivation yields $\alpha_b = \alpha$ and $f_s = f_m^{-(\alpha+1)}$, which for $\alpha \sim -2$ can be approximated $f_s \approx f_m$.

We include best fits for Eqs. (7) and (8) in Table II. As shown in Table II, we find subhalos of a group scale host that appear near the Einstein radius in projection on average retain ∼2% of their infall mass after tidal stripping. Similar to the unevolved case, the scaling term $[F_b(M_h, z)]$ results from the dependence of the total subhalo count and the spatial distribution of subhalos on both halo mass and redshift. We note the spatial distributions differ in the evolved and unevolved cases; therefore we should expect $F_b(M_h, z) \neq F(M_h, z)$.

Equations (7) and (8) can be derived by spatially projecting the Han *et al.* [21] models of spatial and mass distribution into 2D. The Han *et al.* [21] model predicts that the logarithmic slopes of the evolved and unevolved mass functions should be identical, as well as that the logarithmic slopes of the total and projected mass functions should be identical. We check these assumptions against our model results in Table IV. Furthermore, we discuss why this assumption may be inaccurate (at least inaccurate for the $> 10^8 M_\odot$ halos considered here) due to mass segregation from dynamical friction in the unevolved distribution in Sec. IV A. For simplicity, in the remainder of this work we assume $\alpha = \alpha_{\rm b} = -1.97$ to remain consistent with the GALACTICUS results for the evolved distribution (see Table IV).

TABLE III. Amplitude of the evolved PSHMF given in terms of the $f_{\rm s}\cdot\Sigma_{\rm sub}$ [see Eq. (8)]. In the "scaled" column, the number density has been extrapolated to a $10^{13} M_\odot$ halo at redshift $z = 0.5$ using the scaling relations presented in this work. A logarithmic slope of the subhalo mass function $\alpha = -1.97$ was assumed.

Simulation	M_h ($M_\odot$)	z	PSHMF amplitude ($10^{-3}\times$ kpc^{-2})	Scaled ($10^{-3}\times$ kpc^{-2})	Source
GALACTICUS	$1.0\times10^{13.0}$	0.5	11	11	This work
PonosV	$1.2\times10^{13.0}$	0.7	6	5	[58]
PonosQ	$6.5\times10^{12.0}$	0.7	6	6	[58]
Milky Way satellites prior	$1.7\times10^{13.0}$	0.0	1–2	4–8	[60]
Symphony	$1.0\times10^{13.0}$	0.5	5	5	[59]

All dependence on halo mass and redshift in Eqs. (7) and (8) is captured in F and F_b respectively, making $\Sigma_{\rm sub}$ and $f_{\rm s}$ independent of halo mass and redshift. Fits of the Han *et al.* [21] model to the GALACTICUS spatial distribution are given in Table I. Best fits for $\Sigma_{\rm sub}$, $f_{\rm s}$ and α for subhalos within an annulus with inner and outer radii of 10 and 20 kpc are given in Table II. We choose a 10–20 kpc annulus to exclude the innermost subhalo population which may be subject to destruction due to satellite merging implemented in GALACTICUS, as is discussed in Sec. II A 2.

Note that because the evolved projected spatial distribution is nearly independent of the radius near the center of the host, the normalization of the evolved mass (density) function ($f_{\rm s} \cdot \Sigma_{\rm sub}$) will be nearly independent of the choice of aperture radius. We compare $f_{\rm s} \cdot \Sigma_{\rm sub}$ to the PonosV [58] and PonosQ [58] simulations in Table III.

C. Projected subhalo mass function normalization: Impact of central galaxy

Figure 1 shows the impact of the galaxy on the spatial distribution, while Appendix gives more information on the evolution of the central galaxy in our GALACTICUS models. Our GALACTICUS models predict the central galaxy has a minimal impact on the SHMF, as shown in Fig. 4. Within the inner 20 kpc in projection, both the DMO and galactic potential results have no dependence on the radius as shown in Fig. 3. For low mass halos, our models predict the central galaxy introduces a mass-independent rescaling of the SHMF, with no change in slope. The rescaling of the SHMF is dependent on the distance from the host, with the SHMF in the inner region more heavily suppressed when compared to the DMO predictions. For a $10^{13} M_\odot$ halo, the SHMF is suppressed at the $< 5\%$ level over the entire virial

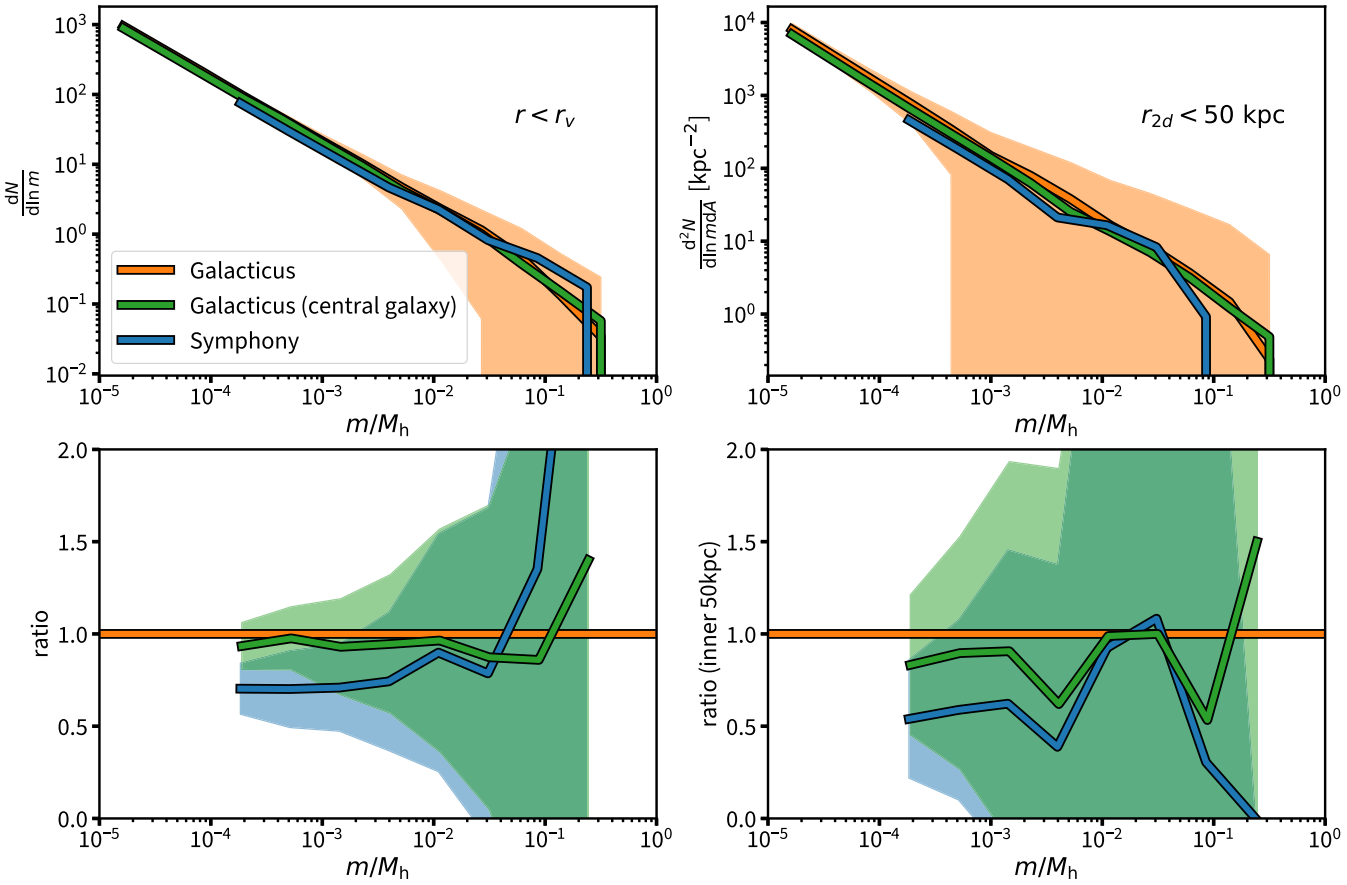


FIG. 4. Upper left: the SHMF over the entire virial volume. Right: the SHMF for subhalos projected within 50 kpc. Results are shown for group mass halos ($M_h \sim 10^{13} M_\odot$) at redshift $z = 0.5$. The GALACTICUS DMO predictions are shown in orange, GALACTICUS predictions for the central galaxy are shown in green, and results for the Symphony suite are shown in blue. Lower left and right show the ratios to the GALACTICUS DMO predictions. The effect of the central galaxy is a reduction in normalization of the SHMF no change to the logarithmic slope. The GALACTICUS and Symphony mass functions and its halo-to-halo scatter are calculated by taking the mean and standard deviation of the value of the mass functions of every halo at every mass bin (in the upper panel, only the GALACTICUS DMO scatter is shown for clarity). For the lower panels, the values of the ratio and its scatter are calculated by dividing the average mass function and its scatter by the average value of the GALACTICUS DMO mass function at each mass bin. Note that the difference in normalization for GALACTICUS with and without a central galaxy is significantly less than the difference between our GALACTICUS models and the Symphony dark matter only simulation suite.

volume when compared to DMO predictions. In the inner 10–20 kpc annulus the suppression increases to ∼15%. We find the impact of the central galaxy is minor when compared to theoretical uncertainties in the SHMF; the difference in normalization between GALACTICUS and Symphony is greater than the difference between GALACTICUS with and without a central galaxy.

D. Projected subhalo mass function scaling: Dark matter only

To parametrize F and F_b we follow the convention of Gilman *et al.* [17] and use the power law expansion

$$F(M_{\rm halo}, z) = \left(\frac{M_{\rm halo}}{10^{13} M_\odot}\right)^{k_1} (z + 0.5)^{k_2} \tag{9}$$

defined to be normalized to 1 at the host halo mass and redshift of a typical lens ($M_{\rm h} = 10^{13} M_\odot$ and $z = 0.5$). A power law is a natural approximation due to the nearly scale-free evolution expected in CDM.

We fit k_1 and k_2 using our suite of GALACTICUS models. We plot the scaling relation in Fig. 5, and give scaling coefficients in Table IV. We find best-fit values of $k_1 = 0.55$ (halo mass scaling coefficient) and $k_2 = 0.37$ (redshift scaling coefficient) for the unevolved distribution and $k_1 = 0.37$ and $k_2 = 1.05$ for the evolved distribution.

The scaling of the PSHMF depends on both the spatial and mass distributions. To model the PSHMF scaling analytically, we project the Han *et al.* [21] model. To scale the normalization of the total SHMF (over the entire virial volume) as a function of the host halo mass and redshift we use the relations provided by Van den Bosch *et al.* [61] for the evolved case and Gao *et al.* [62] for the unevolved case.

We project Eqs. (1) and (5) according to the projection equation

$$\frac{{\rm d}^2 N}{{\rm d}A {\rm d}m}(r_{2d}) = 2 \int_{r_{2d}}^{r_{\rm v}} \frac{u}{\sqrt{u^2 + r_{2d}^2}} \frac{{\rm d}^2 N}{{\rm d}V {\rm d}m}(u) {\rm d}u, \tag{10}$$

assuming a spherically symmetric distribution of subhalos, so that the number density of subhalos depends only on the distance from the center of the host, r. Using the formula provided in Van den Bosch *et al.* [61], the amplitude of the total unevolved SHMF scales as a function of the mass and redshift according to the relation

$$\frac{{\rm d}N}{{\rm d}(m/M_{\rm h})} = C_u \left(\frac{m}{M_{\rm h}}\right)^{\alpha}, \tag{11}$$

where m is the mass of the subhalo (at the time of accretion), $M_{\rm h}$ is the mass of the host halo, α is the logarithmic slope and C_u is the normalization of the unevolved SHMF. For the evolved SHMF, we scale the amplitude using the empirical formula provided by Gao *et al.* [62]:

$$\frac{{\rm d}N}{{\rm d}m_{\rm b}} = C_{\rm e} M_{\rm h} f(m_{\rm b}, z) m_{\rm b}^{\alpha}, \tag{12}$$

where $C_{\rm e}$ is the normalization of the evolved SHMF, $m_{\rm b}$ is the bound mass of the subhalo and $f(m_{\rm b}, z)$ is the abundance of halos with mass $m_{\rm b}$ at redshift z per unit mass in the Universe. We compute $f(M_{\rm h}, z)$ using the Sheth and Tormen [63] mass function.

TABLE IV. The best fits to power law scaling relations predicted by our GALACTICUS models and estimates using the Han *et al.* [21] model. The upper section gives the scaling for the unevolved case, while the lower section gives the scaling results for the evolved case. We compare the scaling relation for the unevolved case for the scaling relation which would be obtained if the scaling was entirely determined by the host's density profile evolution [Eq. (1)]. For the evolved case, we compare the PSHMF scaling prediction to that of the Han *et al.* [21] model [Eq. (5)]. The transfer function from the host's smooth dark matter density profile is parametrized as a power law, with logarithmic slope γ [Eq. (4)]. We consider two possibilities for γ, one where γ is a constant and the other with γ a function of the halo mass. For the case of mass-independent γ, we compute the scaling coefficients for a range of physically meaningful values of γ. Finally, to study the effect of γ varying systematically with mass, we linearly interpolate the best-fit γ values as a function of the mass (Table I) to compute the scaling relations ("γ interp").

Model	k_1 (mass scaling)	k_2 (redshift scaling)
GALACTICUS (unevolved)	0.55	0.37
Host density profile	0.58	0.30
GALACTICUS (evolved)	0.37	1.05
GALACTICUS with central galaxy (evolved)	0.43	1.06
Han *et al.* [21] ($\gamma = 0.8$)	0.37	0.77
Han *et al.* [21] ($\gamma = 1.0$)	0.34	0.86
Han *et al.* [21] ($\gamma = 2.0$)	0.30	1.12
Han *et al.* [21] (γ interp)	0.50	0.83

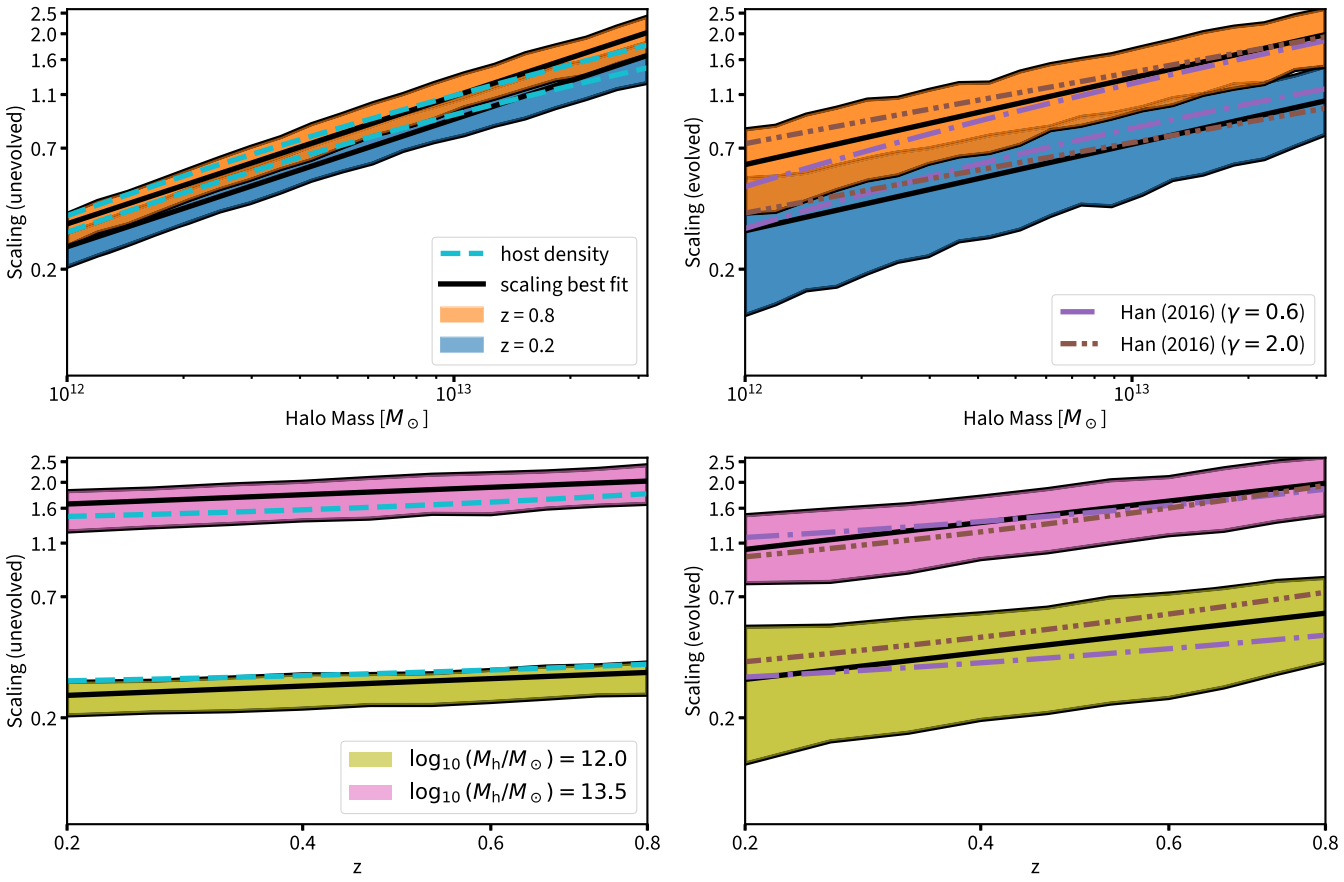


FIG. 5. The scaling shows the ratio of the number of subhalos with mass $8 < \log_{10} m < 9$ projected within an annulus with an inner and outer radius of 10 and 20 kpc respectively to the number of these subhalos in a $10^{13} M_\odot$ host halo at $z = 0.5$ as a function of the host halo mass and redshift. The scaling value and its scatter are calculated by using the mean and standard deviation over all host halos. The left panels show the unevolved subhalo scaling, while the right panel shows the scaling of the evolved distribution. Upper panels show the halo mass scaling relation for host halos at redshift $z = 0.2$ (blue) and $z = 0.8$ (orange). Lower panels show the redshift scaling relation for a host halo mass of $10^{12.0} M_\odot$ (olive) and $10^{13.5} M_\odot$ (pink). Shaded regions show the 1σ halo-to-halo scattered. The unevolved scaling is compared to the scaling of the host's density profile (adjusted to the definition of halo mass used here) in blue (dashed). The scaling relations derived from the Han *et al.* [21] analytic number density distribution [Eqs. (1) and (5)] are shown for $\gamma = 0.6$ (purple, dot dashed) and $\gamma = 2.0$ (brown, dot dashed). These values of γ span a range of physically plausible values.

To calculate the projected scaling relation, we numerically integrate Eq. (5). We first fix γ as constant for all host halo masses, and calculate the projected scaling relations in halo mass and redshift. The range $0.8 \leq \gamma \leq 2.0$ is chosen to represent a reasonable range of possible values of γ, since $\gamma \approx 1$ is expected. The limit as γ goes to 0 gives a Navarro-Frenk-White (NFW) profile (which is the same as the unevolved case). For $\gamma > 3$, the spatial distribution is increasing as a function of r for all values of r, which is nonphysical. In addition to a static value of γ, we consider an additional case where γ varies as a function of the halo mass ("γ interp"). In this case, we linearly interpolate γ as a function of the halo mass, using fits to the spatial distribution predicted by GALACTICUS (see Table I). We summarize our scaling models by providing fits to Eq. (9) in Table II. We also use these scaling relations in Table III to compare our measured values to results from the Aquarius [57] and Phoenix [64] simulations as well as the Milky Way prior used by Nadler *et al.* [60].

E. Projected subhalo mass function scaling: Impact of central galaxy

We find the central galaxy has only a minor impact on the scaling of the amplitude of the PSHMF in redshift and host halo mass (see Table IV). For more massive halos ($M_h > 10^{13} M_\odot$), the difference when compared to the DMO model decreases, which is expected due to the decreasing baryon fraction of the central galaxy (see Fig. 6). Unless otherwise stated, for the remainder of this work we use the DMO scaling relation.

IV. DISCUSSION

A. Subhalo populations of a halo at fixed mass and redshift

Here we discuss the properties of the subhalo populations of group mass ($M_h \sim 10^{13} M_\odot$) halos at redshift $z \sim 0.5$. The GALACTICUS predictions for the unevolved SHMF agree with findings of Van den Bosch *et al.* [61], that the unevolved SHMF can be written in a universal form [Eq. (11)]. Furthermore, we find that GALACTICUS agrees with the empirical formula of Gao *et al.* [62], that for the unevolved case the SHMF scales as a function of the host halo mass and the abundance of halos in the Universe as a whole [Eq. (12)]. Comparing the GALACTICUS and Symphony group halo suite predictions for the normalization of the subhalo mass function within the inner 50 kpc, we find that GALACTICUS predicts on average nearly twice the subhalos in this region. Over the entire virial volume we find similar results, with Symphony predicting around 40% the number of subhalos as GALACTICUS. We note that Symphony and GALACTICUS agree within halo-to-halo scatter, and N-body simulations such as Symphony and Caterpillar disagree on the 25% level [59]. We also find that the suppression of the subhalo mass function due to the potential of the central galaxy is negligible when compared to the current theoretical uncertainty. Similarly, we find that Han *et al.* [21] is in reasonable, although not perfect, agreement with GALACTICUS on the spatial distribution as well as the halo mass and redshift scaling in projection. Here, we provide further discussion for each of these results.

Here, we discuss the expectations for the spatial distributions from the Han *et al.* [21] model and how they compare to predictions from GALACTICUS. The Han *et al.* [21] model describes the unevolved spatial distribution of subhalos as following the dark matter density distribution of the host halo. As shown in Fig. 2, our GALACTICUS models show a difference between the unevolved subhalo spatial distribution and the host's smooth dark matter density profile (normalized to 1 at $r = r_\mathrm{v}$).

At first glance, this is difficult to explain with dynamical friction, as the dynamical friction timescale is proportional to the subhalo-to-host-halo mass ratio (m/M_h) and is longer than the age of the Universe for subhalos with with less than $1/20$ the mass of the host halo [65]. However, older subhalos will have fallen into a much less massive host (in which m/M_h is much closer to 1), and therefore they would experience a much smaller dynamical friction timescale.

To determine if dynamical friction is the cause of this difference, GALACTICUS could be run with and without dynamical friction enabled. We leave this to future work. The Han *et al.* [21] model does not include effects of dynamical friction, while the GALACTICUS models do. For the evolved case, Han *et al.* [21] use a power law profile to model the ratio of evolved spatial distribution to the host's dark matter density profile. Figure 2 shows that GALACTICUS does not follow this expectation exactly, with deviations from a power law near the virial radius.

When comparing our GALACTICUS models to the Symphony Group suite at redshift $z = 0.5$, we find that GALACTICUS predicts a normalization of the projected SHMF a factor of $\sim$2 times higher than Symphony. We note that this difference remains within 1σ halo-to-halo scatter and caution should be taken when interpreting this result due to the small number of subhalos in this volume which makes obtaining good statistics challenging. If the difference is statistically significant, there are several possible explanations for the discrepancy. A large body of work has drawn attention to artificial disruption present in N-body simulations [27–30]. An estimate for spurious suppression due to artificial disruption is 10%–20% over the entire virial volume, only increasing to a maximum factor of 3 within the inner 2% of the virial radius [30]. Another possible explanation is inaccuracies in halo finding algorithms. In our work, we use results based on the Symphony subhalo catalogs extracted using the Rockstar [66] halo finder. Recent work using a new halo finding algorithm by Mansfield *et al.* [67] has identified 15%–40% more subhalos within the virial radius when compared to the Rockstar results, increasing to 35%–120% within $r < r_\mathrm{v}/4$. Another possibility may be the GALACTICUS physics models. Our GALACTICUS model does not include a model of preinfall tidal stripping [68]. We do not make a determination if the discrepancy is numerical in nature and leave studying the underlying cause(s) of the differences to future work. We find subhalos of a group scale host that appear near the Einstein radius in projection on average retain $\sim$4% of their infall mass after tidal stripping.

B. Subhalo population dependence on host properties

We discuss the scaling the projected SHMF as a function of the halo mass and redshift. For the evolved case, our scaling relations from GALACTICUS and our analytic models predict that for a factor of 10 increase in host halo mass the evolved PSHMF function will increase by a factor $\sim$2. This is similar to the factor of $\sim$3 used in the extrapolation performed in Xu *et al.* [24], but differs from the factor of $\approx$8 used Gilman *et al.* [18]. Differences from Gilman *et al.* [18] are likely due to a change in GALACTICUS, in particular an update to the treatment of higher order substructure. The change in higher order substructure treatment had a large impact on the spatial distributions, with the previous spatial distributions of subhalos being more cuspy.

Next, we compare our scaling results from GALACTICUS to the Han *et al.* [21] model of the spatial transfer functions. Three forms of the tidal stripping transfer function $\hat{T}$ are considered [see Eqs. (5) and (12)], with fits to the projected scaling relation fits provided in Table IV. All cases considered result in host halo mass scaling similar to GALACTICUS, with a 10 times increase in mass resulting

in a roughly ≈2 times increase in the normalization of the PSHMF. Fixing $\hat{T}(r/r_v)$ [see Eqs. (4) and (5)] to be invariant in the halo mass ($\gamma = 0.6$, 1.0, 2.0 and "GALACTICUS interp" cases in Table IV) results in a best fit of $k_1 \approx 0.3$. The redshift scaling results (k_2) range from 0.83 to 1.12, which is in agreement with the $k_2 = 1.05$ predicted by GALACTICUS. Allowing γ to vary with the halo mass (linearly interpolating between $\gamma = 1.06$ and $\gamma = 0.67$) does not increase agreement with the GALACTICUS models.

C. Impact of the central galaxy

Finally, we discuss the impacts of the central galaxy. We find that the central galaxy impacts the subhalo mass function normalization on the less than 5% level over the entire virial volume and the less than 15% level within a projected aperture near the Einstein radius. We note that studies of the effects of an analytic potential on the subhalos in Milky-Way-like systems have found a more significant reduction in the subhalo mass function than what we have found here [69,70]. Webb and Bovy [70] simulate individual dark matter only subhalos in evolving external tidal fields with and without baryonic disk and bulge components finding that the inclusion of these baryonic components reduces the number of subhalos by 65%. We highlight two possible reasons why the impact of a group scale massive central elliptical may be less significant than that of a Milky-Way-like central galaxy. First, the ratio of central galaxy mass to halo mass is, on average, nearly a factor of 2 lower at the group scale [36]. Second, the systems are physically distinct: The Milky Way contains a disk component which is not found in massive ellipticals.

D. Recommendations for future $\Sigma_{\rm sub}$ priors

Here, we discuss our recommendations for future priors on the number density of subhalos within the Einstein radius. To ensure matching of observable quantities, we use our GALACTICUS models to recommend a prior on the amplitude of the evolved halo mass function $f_s \cdot \Sigma_{\rm sub}$, instead of $\Sigma_{\rm sub}$ directly. Our GALACTICUS models predict an average evolved number density of $f_s \cdot \Sigma_{\rm sub} \sim 10^{-2}$ kpc^{-2}, while the Symphony suite predicts $f_s \cdot \Sigma_{\rm sub} \sim 5 \times 10^{-3}$ kpc^{-2} (additional results are given in Table III). Here, we do not investigate the possible reasons for the differences, and instead treat each result as equally likely so we can estimate a theoretical uncertainty. Across works compared here, we find that the value of $f_s \cdot \Sigma_{\rm sub}$ varies from 10^{-3} to 10^{-2} kpc^{-2}, spanning approximately an order of magnitude, with more recent works in the $5 \times 10^{-3} - 10^{-2}$ kpc^{-2} range. In a 100 kpc^2 aperture, this approximately corresponds to 0.1 to 1 subhalos with bound mass $8 < \log_{10}(m/M_\odot) \leq 9$ (or equivalently a number density of $10^{-3} - 10^{-2}$ kpc^2). For reference, an average Einstein radius of 1 arc sec at redshift $z = 0.5$ corresponds to an ∼150 kpc^2 aperture. We note that this is well within the allotted uncertainty of previous lensing studies. Furthermore, we estimate that the impact of a central galaxy (reduction in $f_s \cdot \Sigma_{\rm sub}$ by a factor of ∼10%) is negligible compared to the current theoretical uncertainty between different models.

We recommend a prior centered around $f_s \cdot \Sigma_{\rm sub} = 10^{-2}$ kpc^{-2} and allowing for at least a factor of 2 (∼0.3 dex) uncertainty. We note that this is a similar prior to the tighter prior considered by Gilman *et al.* [26].

When Gilman *et al.* [26] compared dark matter inferences using a 1.5 dex log-uniform prior (similar to that used in previous lensing studies) with an inference using a tighter Gaussian prior with width of 0.2 dex, the lower bound on the half-mode mass improved by 2 dex. We recommend the continued use of a log-uniform prior as N-body simulations have consistently lower values of $f_s \cdot \Sigma_{\rm sub}$ when compared to our GALACTICUS models; a log-uniform prior treats both cases as equally likely. By implementing a 0.3 dex log-uniform prior we expect similar improvements on constraining power to those observed by Gilman *et al.* [26].

V. SUMMARY

In this work we study predictions for the population level statistics of subhalos of group mass halos at small projected distances. Particular focus is placed on the normalization of the projected SHMF, where we give recommendations for a more informed prior on this quantity than previous lensing studies. Here we provide a summary of key results in our paper:

(i) We present a new suite of GALACTICUS models of the substructure of host halos with masses $12 \leq \log_{10}(M_h/M_\odot) \leq 13.5$ and redshifts $0.2 \leq z \leq 0.8$.

(ii) Using these simulations, we measure the projected number density of subhalos near the Einstein radius. The scaling in halo mass and redshift is well described with a power law with coefficients $k_1 = 0.55$ (halo mass) and $k_2 = 0.37$ (redshift) for the unevolved case and $k_1 = 0.37$ and $k_2 = 1.05$ for the evolved case.

The projected scaling differs from the scaling of the total number of satellites within the virial volume (total subhalo count scales nearly linearly with the mass of the host and is almost independent of redshift in both the evolved and unevolved cases).

(iii) The evolved and unevolved projected SHMF amplitude scales differently as a function of the halo mass and redshift due to different spatial distributions between the two cases. For the evolved case, the projected SHMF amplitude approximately scales with the cube root of the host halos mass and nearly linearly in redshift. We find the scaling of the

PSHMF in redshift and mass to be in excellent agreement between simulations and analytic models.

Our analytic models show the scaling of the PSHMF can be explained in terms of two factors: (1) the scaling of the total number of subhalos within the virial volume of the host and (2) the spatial distribution of subhalos.

(iv) On the group scale, a central galaxy reduces the normalization of the evolved SHMF by $<5\%$ over the entire virial volume, increasing to 15% for the PSHMF within the inner 20 kpc. This is much less than the current theoretical uncertainty between different models/N-body simulations.

For future lensing studies with group mass lenses, we find that the effects from a central galaxy have a negligible impact on the strength of tidal stripping experienced by subhalos and does not need to be considered when setting a prior $\Sigma_{\rm sub}$.

(v) We find subhalos of a group scale host that appear near the Einstein radius in projection on average retain $\sim 2\%$ of their infall mass after tidal stripping.

(vi) We compare the projected number density of subhalos in a variety of N-body simulations to our GALACTICUS models. We find that all models/simulations considered agree on the projected SHMF normalization well within a factor of ~ 2. Additionally, we find that GALACTICUS and Symphony agree within the 1σ halo-to-halo scatter for subhalos halos in the bound mass range $9 < \log_{10}(m_b/M_\odot) \leq 10$.

We recommend using a stronger prior on $f_{\rm s} \cdot \Sigma_{\rm sub}$ when compared to previous lensing studies; specifically a log-uniform prior centered around $f_s \cdot \Sigma_{\rm sub} = 10^{-2}$ kpc^{-2} and allowing for at least a factor of 2 uncertainty. Compared to previous lensing studies, we expect a significant reduction in uncertainty on parameters like half-mode mass when using this tighter prior.

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DATA AVAILABILITY

The data that support the findings of this article are openly available [71–73].

APPENDIX: EVOLUTION OF THE CENTRAL GALAXY

In our models with a central galaxy, we model the evolution of the potential of a central massive elliptical using empirical relations from Behroozi *et al.* [36]. We plot properties of the central galaxy in our GALACTICUS models in Fig. 6.

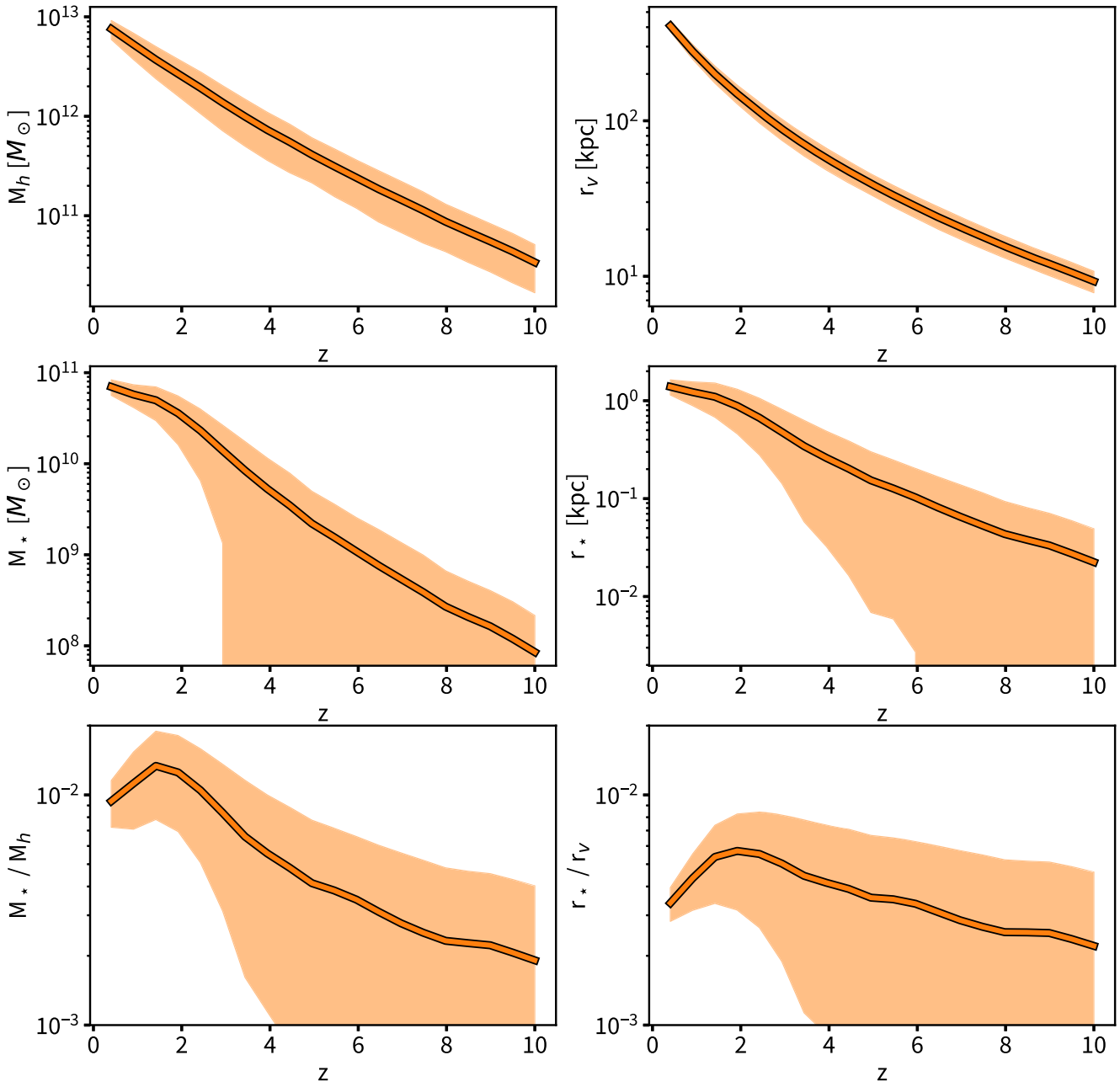


FIG. 6. Our GALACTICUS model predictions for the evolution of a $10^{13}M_\odot$ halo and its central galaxy. Stellar masses are determined using the Behroozi *et al.* [36] relation between halo mass and stellar mass, while galactic radii are determined using empirical fits to the stellar mass to stellar radius relation provided by Shen *et al.* [55]. For simplicity, no scatter has been included in either of these relationships. The top left panel gives the evolution of the mass of the central halo while the top right panel gives the evolution of the virial radius of the halo. The middle left panel shows the evolution of the stellar mass, while the middle right panel shows the evolution of the Hernquist radius of the central galaxy. Finally, the lower left and right panels show the evolution of the ratio of stellar mass to halo mass and Hernquist radius to virial radius, respectively. The curve shows the mean value, while the shaded region shows 1σ scatter.

[1] Planck Collaboration, Planck2018 results: VI. Cosmological parameters, Astron. Astrophys. **641**, A6 (2020).

[2] S. D. White and M. J. Rees, Core condensation in heavy halos: A two-stage theory for galaxy formation and clustering, Mon. Not. R. Astron. Soc. **183**, 341 (1978).

[3] S. D. White and C. S. Frenk, Galaxy formation through hierarchical clustering, Astrophys. J. **379**, 52 (1991).

[4] W. De Blok, F. Walter, E. Brinks, C. Trachternach, S. Oh, and R. C. Kennicutt, High-resolution rotation curves and galaxy mass models from things, Astron. J. **136**, 2648 (2008).

[5] D. H. Weinberg, J. S. Bullock, F. Governato, R. Kuzio de Naray, and A. H. Peter, Cold dark matter: Controversies on small scales, Proc. Natl. Acad. Sci. U.S.A. **112**, 12249 (2015).

[6] A. Bonaca, D. W. Hogg, A. M. Price-Whelan, and C. Conroy, The Spur and the gap in GD-1: Dynamical evidence for a dark substructure in the Milky Way halo, Astrophys. J. **880**, 38 (2019).

[7] N. Banik, J. Bovy, G. Bertone, D. Erkal, and T. J. L. de Boer, Evidence of a population of dark subhaloes from gaia and Pan-STARRS observations of the GD-1 stream, Mon. Not. R. Astron. Soc. **502**, 2364 (2021).

[8] N. Banik, J. Bovy, G. Bertone, D. Erkal, and T. De Boer, Novel constraints on the particle nature of dark matter from stellar streams, J. Cosmol. Astropart. Phys. 10 (2021) 043.

[9] J. Bovy, D. Erkal, and J. L. Sanders, Linear perturbation theory for tidal streams and the small-scale CDM power spectrum, Mon. Not. R. Astron. Soc. **466**, 628 (2017).

[10] N. Banik, G. Bertone, J. Bovy, and N. Bozorgnia, Probing the nature of dark matter particles with stellar streams, J. Cosmol. Astropart. Phys. 07 (2018) 061.

[11] S. Vegetti, G. Despali, M. Lovell, and W. Enzi, Constraining sterile neutrino cosmologies with strong gravitational lensing observations at redshift z 0.2, Mon. Not. R. Astron. Soc. **481**, 3661 (2018).

[12] Q. Minor, S. Gad-Nasr, M. Kaplinghat, and S. Vegetti, An unexpected high concentration for the dark substructure in the gravitational lens SDSSJ0946+ 1006, Mon. Not. R. Astron. Soc. **507**, 1662 (2021).

[13] D. M. Powell, S. Vegetti, J. McKean, S. D. White, E. G. Ferreira, S. May, and C. Spingola, A lensed radio jet at milli-arcsecond resolution-II. Constraints on fuzzy dark matter from an extended gravitational arc, Mon. Not. R. Astron. Soc. **524**, L84 (2023).

[14] S. Mao and P. Schneider, Evidence for substructure in lens galaxies?, Mon. Not. R. Astron. Soc. **295**, 587 (1998).

[15] R. E. Keeley, A. M. Nierenberg, D. Gilman, C. Gannon, S. Birrer, T. Treu, A. J. Benson, X. Du, K. Abazajian, T. Anguita *et al.*, JWST lensed quasar dark matter survey II: Strongest gravitational lensing limit on the dark matter free streaming length to date, Mon. Not. R. Astron. Soc. **535**, 1652 (2024).

[16] V. Dike, D. Gilman, and T. Treu, Strong lensing constraints on primordial black holes as a dark matter candidate, Mon. Not. R. Astron. Soc. **522**, 5434 (2023).

[17] D. Gilman, S. Birrer, T. Treu, A. Nierenberg, and A. Benson, Probing dark matter structure down to 107 solar masses: flux ratio statistics in gravitational lenses with line-of-sight haloes, Mon. Not. R. Astron. Soc. **487**, 5721 (2019).

[18] D. Gilman, S. Birrer, A. Nierenberg, T. Treu, X. Du, and A. Benson, Warm dark matter chills out: Constraints on the halo mass function and the free-streaming length of dark matter with eight quadruple-image strong gravitational lenses, Mon. Not. R. Astron. Soc. **491**, 6077 (2019).

[19] D. Gilman, A. Benson, J. Bovy, S. Birrer, T. Treu, and A. Nierenberg, The primordial matter power spectrum on subgalactic scales, Mon. Not. R. Astron. Soc. **512**, 3163 (2022).

[20] D. Gilman, Y.-M. Zhong, and J. Bovy, Constraining resonant dark matter self-interactions with strong gravitational lenses, Phys. Rev. D **107**, 103008 (2023).

[21] J. Han, S. Cole, C. S. Frenk, and Y. Jing, A unified model for the spatial and mass distribution of subhaloes, Mon. Not. R. Astron. Soc. **457**, 1208 (2016).

[22] R. Errani and J. F. Navarro, The asymptotic tidal remnants of cold dark matter subhaloes, Mon. Not. R. Astron. Soc. **505**, 18 (2021).

[23] X. Du, D. Gilman, T. Treu, A. Benson, and C. Gannon, Faster than SAM: An empirical model for the tidal evolution of dark matter substructure around strong gravitational lenses, arXiv:2503.07728.

[24] D. Xu, D. Sluse, L. Gao, J. Wang, C. Frenk, S. Mao, P. Schneider, and V. Springel, How well can cold dark matter substructures account for the observed radio flux-ratio anomalies, Mon. Not. R. Astron. Soc. **447**, 3189 (2015).

[25] G. Despali, S. Vegetti, S. D. M. White, C. Giocoli, and F. C. van den Bosch, Modelling the line-of-sight contribution in substructure lensing, Mon. Not. R. Astron. Soc. **475**, 5424 (2018).

[26] D. Gilman, S. Birrer, A. Nierenberg, and M. S. Oh, Turbocharging constraints on dark matter substructure through a synthesis of strong lensing flux ratios and extended lensed arcs, Mon. Not. R. Astron. Soc. **533**, 1687 (2024).

[27] F. C. van den Bosch, G. Ogiya, O. Hahn, and A. Burkert, Disruption of dark matter substructure: fact or fiction?, Mon. Not. R. Astron. Soc. **474**, 3043 (2017).

[28] S. B. Green and F. C. van den Bosch, The tidal evolution of dark matter substructure—I. Subhalo density profiles, Mon. Not. R. Astron. Soc. **490**, 2091 (2019).

[29] R. Errani and J. Peñarrubia, Can tides disrupt cold dark matter subhaloes?, Mon. Not. R. Astron. Soc. **491**, 4591 (2019).

[30] A. J. Benson and X. Du, Tidal tracks and artificial disruption of cold dark matter haloes, Mon. Not. R. Astron. Soc. **517**, 1398 (2022).

[31] B. F. Griffen, A. P. Ji, G. A. Dooley, F. A. Gómez, M. Vogelsberger, B. W. O'Shea, and A. Frebel, The caterpillar project: A large suite of Milky Way sized halos, Astrophys. J. **818**, 10 (2016).

[32] R. Errani and J. Peñarrubia, Can tides disrupt cold dark matter subhaloes?, Mon. Not. R. Astron. Soc. **491**, 4591 (2020).

[33] X. Du, A. Benson, Z. Carton Zeng, T. Treu, A. H. G. Peter, C. Mace, F. Jiang, S. Yang, C. Gannon, D. Gilman, A. M. Nierenberg, and E. O. Nadler, Tidal evolution of cored and cuspy dark matter halos, Phys. Rev. D **110**, 023019 (2024).

[34] A. J. Benson, GALACTICUS: A semi-analytic model of galaxy formation, New Astron. **17**, 175 (2012).

[35] Y. Shu, J. R. Brownstein, A. S. Bolton, L. V. Koopmans, T. Treu, A. D. Montero-Dorta, M. W. Auger, O. Czoske, R. Gavazzi, P. J. Marshall *et al.*, The Sloan Lens ACS survey. XIII. Discovery of 40 new galaxy-scale strong lenses, Astrophys. J. **851**, 48 (2017).

[36] P. Behroozi, R. H. Wechsler, A. P. Hearin, and C. Conroy, UNIVERSEMACHINE: The correlation between galaxy growth and dark matter halo assembly from $z = 0$–10, Mon. Not. R. Astron. Soc. **488**, 3143 (2019).

[37] ASTROPY Collaboration, ASTROPY: A community Python package for astronomy, Astron. Astrophys. **558**, A33 (2013).

[38] ASTROPY Collaboration, The ASTROPY Project: Building an open-science Project and Status of the v2.0 Core Package, Astron. J. **156**, 123 (2018).

[39] ASTROPY Collaboration, The ASTROPY Project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the Core Package, Astrophys. J. **935**, 167 (2022).

[40] B. Diemer, COLOSSUS: A Pythontoolkit for cosmology, large-scale structure, and dark matter halos, Astrophys. J. Suppl. Ser. **239**, 35 (2018).

[41] B. Diemer and M. Joyce, An accurate physical model for halo concentrations, Astrophys. J. **871**, 168 (2019).

[42] S. Cole, C. G. Lacey, C. M. Baugh, and C. S. Frenk, Hierarchical galaxy formation, Mon. Not. R. Astron. Soc. **319**, 168 (2000).

[43] H. Parkinson, S. Cole, and J. Helly, Generating dark matter halo merger trees, Mon. Not. R. Astron. Soc. **383**, 557 (2008).

[44] J. Bond, S. Cole, G. Efstathiou, and N. Kaiser, Excursion set mass functions for hierarchical Gaussian fluctuations, Astrophys. J. **379**, 440 (1991).

[45] R. G. Bower, The evolution of groups of galaxies in the Press-Schechter formalism, Mon. Not. R. Astron. Soc. **248**, 332 (1991).

[46] W. H. Press and P. Schechter, Formation of galaxies and clusters of galaxies by self-similar gravitational condensation, Astrophys. J. **187**, 425 (1974).

[47] L. Jiang, S. Cole, T. Sawala, and C. S. Frenk, Orbital parameters of infalling satellite haloes in the hierarchical λCDM model, Mon. Not. R. Astron. Soc. **448**, 1674 (2015).

[48] S. Chandrasekhar, Dynamical friction. I. General considerations: The coefficient of dynamical friction, Astrophys. J. **97**, 255 (1943).

[49] A. R. Zentner, A. A. Berlind, J. S. Bullock, A. V. Kravtsov, and R. H. Wechsler, The physics of galaxy clustering. I. A model for subhalo populations, Astrophys. J. **624**, 505 (2005).

[50] O. Y. Gnedin, L. Hernquist, and J. P. Ostriker, Tidal shocking by extended mass distributions, Astrophys. J. **514**, 109 (1999).

[51] S. Yang, X. Du, A. J. Benson, A. R. Pullen, and A. H. G. Peter, A new calibration method of sub-halo orbital evolution for semi-analytic models, Mon. Not. R. Astron. Soc. **498**, 3902 (2020).

[52] F. C. van den Bosch and G. Ogiya, Dark matter substructure in numerical simulations: A tale of discreteness noise, runaway instabilities, and artificial disruption, Mon. Not. R. Astron. Soc. **475**, 4066 (2018).

[53] O. Y. Gnedin, A. V. Kravtsov, A. A. Klypin, and D. Nagai, Response of dark matter halos to condensation of baryons: Cosmological simulations and improved adiabatic contraction model, Astrophys. J. **616**, 16 (2004).

[54] L. Hernquist, An analytical model for spherical galaxies and bulges, Astrophys. J. **356**, 359 (1990).

[55] S. Shen, H. J. Mo, S. D. M. White, M. R. Blanton, G. Kauffmann, W. Voges, J. Brinkmann, and I. Csabai, The size distribution of galaxies in the Sloan Digital Sky Survey, Mon. Not. R. Astron. Soc. **343**, 978 (2003).

[56] D. G. York, J. Adelman, J. E. Anderson Jr, S. F. Anderson, J. Annis, N. A. Bahcall, J. Bakken, R. Barkhouser, S. Bastian, E. Berman *et al.*, The sloan digital sky survey: Technical summary, Astron. J. **120**, 1579 (2000).

[57] V. Springel, J. Wang, M. Vogelsberger, A. Ludlow, A. Jenkins, A. Helmi, J. F. Navarro, C. S. Frenk, and S. D. M. White, The Aquarius Project: The subhaloes of galactic haloes, Mon. Not. R. Astron. Soc. **391**, 1685 (2008).

[58] D. Fiacconi, P. Madau, D. Potter, and J. Stadel, Cold dark matter substructures in early-type galaxy halos, Astrophys. J. **824**, 144 (2016).

[59] E. O. Nadler, P. Mansfield, Y. Wang, X. Du, S. Adhikari, A. Banerjee, A. Benson, E. Darragh-Ford, Y.-Y. Mao, S. Wagner-Carena *et al.*, Symphony: Cosmological zoom-in simulation suites over four decades of host halo mass, Astrophys. J. **945**, 159 (2023).

[60] E. O. Nadler, S. Birrer, D. Gilman, R. H. Wechsler, X. Du, A. Benson, A. M. Nierenberg, and T. Treu, Dark matter constraints from a unified analysis of strong gravitational lenses and Milky Way satellite galaxies, Astrophys. J. **917**, 7 (2021).

[61] F. C. Van den Bosch, G. Tormen, and C. Giocoli, The mass function and average mass-loss rate of dark matter subhaloes, Mon. Not. R. Astron. Soc. **359**, 1029 (2005).

[62] L. Gao, S. D. M. White, A. Jenkins, F. Stoehr, and V. Springel, The subhalo populations of ΛCDM dark haloes, Mon. Not. R. Astron. Soc. **355**, 819 (2004).

[63] R. K. Sheth and G. Tormen, Large-scale bias and the peak background split, Mon. Not. R. Astron. Soc. **308**, 119 (1999).

[64] L. Gao, J. F. Navarro, C. S. Frenk, A. Jenkins, V. Springel, and S. D. M. White, The Phoenix project: The dark side of rich Galaxy clusters, Mon. Not. R. Astron. Soc. **425**, 2169 (2012).

[65] M. Boylan-Kolchin, C.-P. Ma, and E. Quataert, Dynamical friction and galaxy merging time-scales, Mon. Not. R. Astron. Soc. **383**, 93 (2008).

[66] P. S. Behroozi, R. H. Wechsler, and H.-Y. Wu, The rockstar phase-space temporal halo finder and the velocity offsets of cluster cores, Astrophys. J. **762**, 109 (2012).

[67] P. Mansfield, E. Darragh-Ford, Y. Wang, E. O. Nadler, B. Diemer, and R. H. Wechsler, Symfind: Addressing the fragility of subhalo finders and revealing the durability of subhalos, Astrophys. J. **970**, 178 (2024).

[68] P. S. Behroozi, R. H. Wechsler, Y. Lu, O. Hahn, M. T. Busha, A. Klypin, and J. R. Primack, Mergers and mass accretion for infalling halos both end well outside cluster virial radii, Astrophys. J. **787**, 156 (2014).

[69] E. D'Onghia, V. Springel, L. Hernquist, and D. Keres, Substructure depletion in the Milky Way halo by the disk, Astrophys. J. **709**, 1138 (2010).

[70] J. J. Webb and J. Bovy, High-resolution simulations of dark matter subhalo disruption in a Milky-Way-like tidal field, Mon. Not. R. Astron. Soc. **499**, 116 (2020).

[71] All raw GALACTICUS tree data are available at https://ucmerced.box.com/s/gl3lg60e80783m0hq3po4zrzgi9t63bt.

[72] Summaries of GALACTICUS suites are available at https://ucmerced.box.com/s/ugtvs2rto2x5opbv2mlesbdnze2un87m.

[73] Github repository used to analyze the data is available at https://github.com/cgannonucm/lpanalysis/tree/gannon2025-revision-2.